MILITARY SPECIFICATION

LAMPHOLDER, PAR 46/64, AVIATION LIGHTING

This specification has been approved by the Bureau of Naval Weapons Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers the requirements for PAR 46 and PAR 64 lampholders utilizing lamps with screw terminal ends. Lampholders shall have a fixed beam and shall be furnished with lamp ratings up to 500 watts, as specified (see 6.2 and 3.4.1).

1.2 Classification - Lampholders shall be of the following types, as specified (see 6.2):

Type I - For other Navy applications - (PAR 46-200W lamp)

Type II - Wheels-up warning system - (PAR 64-500W lamp)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-S-763	Steel Bars, Shapes, and Forgings-Corrosion Resisting
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood

FSC 6210

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SPECIFICATIONS

Federal (Continued)

PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-B-665	Boxes, Paperboard, Metal Stayed (Including Stay Material)
PPP-B-676	Boxes, Set-Up, Paperboard
PPP-T-60	Tape: Pressure Sensitive Adhesive, Waterproof for Packaging and Sealing

Military

MIL-W-76	Wire and Cable, Hookup-Electrical-Insulated
JAN-P-100	Packaging and Packing for Overseas Shipment, General Specification
MIL-P-116	Preservation, Methods of
MIL-B-138	Boxes, Wood, Fiberboard-Lined for Overseas Shipment (For Weight Contents Not Exceeding 500 Pounds)
MIL-C-915	Cable, Cord and Wire, Electrical Shipboard Use
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-E-5556	Enamel, Camouflage, Quick Drying
MIL-E-7729	Enamel, Gloss
MIL-C-7989	Covers, Light-Transmitting, for Aeronautical Lights, General Specification for

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SPECIFICATIONS

Military (Continued)

MIL-P-8585	Primer Coating Zinc Chromate, Low-Moisture-sensitivity
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-B-10377	Box, Wood, Cleated, Veneer, Paper Overlaid
MIL-L-10547	Liners, Case, Waterproof
MIL-C-25050	Colors, Aeronautical Lights and Lighting Equipment, General Requirements for
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-9	Screw Thread Conventions and Methods of Specifying
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MS25036	Terminal Lug, Crimp Style, Copper , Insulated, Class 1
MS33586	Metals, Definition of Dissimilar
MS35440	Terminal Lug, Solder Type, Copper Stamping, Insulation Grip, Slotted
MS90303	Lampholders, Par 46 and Par 64, Aviation Lighting

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STANDARDS

Military (Continued)

MS24489	Filter, Marker Light, Airport Approach, Glass
MS24502	Filter, Marker Light, Airport Runway, Glass
MS25241-4553	Lamp, Incandescent, Par-46 Bulb, Screw Terminal Base, Landing
MS25242-4552	Lamp, Incandescent, Par-64 Bulb, Screw Terminal Base

DRAWINGS

BuShips Dwg No. 9000-S6505-B-73214-68	Terminal Block

(When requesting any of the applicable documents refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania, Attention Code DSI.)

2.2 Other publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

Federal Aviation Agency

L-823	Plug and Receptacle Cable Connectors

(Copies of FAA documents may be obtained from the Federal Aviation Agency, Washington 25, D. C.)

National Bureau of Standards

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

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Consolidated Classification Committee

Uniform Freight Classification Rules

(Application for copies of the above publication should be made to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

3. REQUIREMENTS

3.1 Preproduction - The lampholder furnished under this specification shall be a product which has been tested and passed the preproduction inspection specified herein.

3.2 Selection of specifications and standards - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with Standard MIL-STD-143, except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial parts - Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, and cotter pins having suitable properties may be used provided:

(a) They can be replaced by the standard parts (MS) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.2.2 Standard parts - With the exception in 3.2.1, MS standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials -

3.3.1 Fungus-proof materials - Materials that are nutrients for fungi shall not be used where it is practicable to avoid them. When used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

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3.3.2 Metals - Metals shall be of the corrosion-resistant type or suitable treated to resist corrosion caused by fuels, salt spray, or atmospheric conditions as may be encountered in storage or normal service. The use of dissimilar metals shall be avoided whenever practicable. Dissimilar metals are defined in Standard MS33586.

3.3.3 Bolts, screws, and nuts - Bolts, screws, and nuts shall be of corrosion-resistant metal or shall be treated to prevent corrosion or seizing which would cause difficulty in removal for servicing and maintenance.

3.3.4 Lenses - Any lenses utilized in the light shall be made of class B glass conforming to Specification MIL-C-7989.

3.4 Design and construction - The lampholder shall be so designed and constructed that no parts will work loose in service. The lampholder shall withstand the strains, jars, vibration, and other conditions incident to shipping, storage, installation, and service.

3.4.1 Lamps - The lampholder shall be designed for use with screw terminal ends, sealed beam lamps, both 500W Par 64 and 200W Par 46. Unless otherwise specified, lamps shall not be shipped with the lampholder (see 6.2). There shall be approximately a 1-inch space between the lamp and the filter.

3.4.2 Design approval - Unless otherwise specified, the lampholder shall conform to the configuration of MS90303. Slightly different details of construction will be satisfactory, provided the difference does not affect the performance or interchangeability of the lampholders, and provided the design is approved by the procuring activity. Lampholder designs shall be submitted to the procuring activity for approval prior to fabrication.

3.4.3 Design conditions - All lampholders shall be designed for continuous outdoor operation under all weather conditions.

3.4.4 Adjustment and repairs - The lampholder shall be constructed so that adjustments and repairs can be easily made by personnel of operating units and overhaul bases without the use of special tools.

3.5 Components - The major components of the lampholder shall consist of the following: Housing shell, ring, support and holder, terminal block, clips, and connectors. (Lamps and filters are not usually furnished with the lights.)

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3.5.1 Housing Shell - The housing shell shall be made in accordance with MS90303. The aluminum casting, however, shall be acceptable only if interchangeable with the aluminum sheet shells. If cast aluminum shell is provided, interior and exterior surface shall be smooth and free of pits and mold marks. Spun shells will not be acceptable.

3.5.2 Ring - Unless otherwise specified, the ring shall be of type in conformance with MS90303.

3.5.3 Support and holder - An adjustable mount shall be furnished consisting of a holder and a support. Both items shall be in accordance with MS90303.

3.5.3.1 Mount hardware - The screws, nuts, bolts, washers, etc., used for connecting the adjustable mount together and to the shell shall be in accordance with MS90303.

3.5.4 Terminal block - The terminal block shall be of the type in conformance with MS90303.

3.5.5 Lamp clips - Three pressure clips shall be provided to hold the lamp to the index ring. The clips shall be in accordance with MS90303. They shall be constructed so that they will not cut into the lamp and will permit the lamp to be readily clipped and held in place. Stainless steel screws, nuts, and washers shall be furnished with the clips.

3.5.6 Cord - Cord size will be specified on the procurement document (see 6.2).

3.5.6.1 Cord plug - The plug shall be a 2-pole, 600V, 20-amp plug in accordance with type 1a of FAA Specification L-823 and shall be molded to the lower end of the cord.

3.5.6.2 The cord shall be threaded through the mounting support so that the cord and its connector are inside the mounting support.

3.5.7 Squeeze connectors - The squeeze connectors shall be in accordance with MS90303.

3.5.8 Filter - The filters shall be in accordance with MS24489, MS24502 and MIL-C-7989. Unless otherwise specified, filters shall not be furnished with the light (see 6.2).

3.5.8.1 Filter clips - Three filter holder clips shall be provided for mounting the filter. They shall be in accordance with MS90303. Stainless steel screws, nuts, and washers shall be furnished with the clips.

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3.5.8.2 Filter color - The filter color shall be as specified in MIL-L-25050 and MIL-C-7989 (see 6.2).

3.6 Performance -

3.6.1 Operation - There shall be no evidence of smoking, charring, softening, or other damage to any components of the light when subjected to the test of 4.5.3.

3.6.2 Dielectric - When tested in accordance with 4.5.6, the insulation resistance of the terminal block shall be 2,000 megohms or greater.

3.6.3 Color and transmission - The filter shall conform to the requirements for color and transmission of Specification MIL-C-25050 when tested as specified in 4.5.4.

3.6.4 Lamp position - When tested as specified in 4.5.5, the aiming of the light shall correspond to the markings on the protractor scale. (See sheet 6 of MS90303).

3.7 Interchangeability - All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of Specification MIL-D-70327 shall govern manufacturer's part numbers and changes thereto.

3.8 Finish - All aluminum parts, except the inside of the support, shall be painted in accordance with MS. All paints and finishes used in the lampholder shall be suitable for the operating temperatures involved and shall not cause parts of the assembly to adhere to each other.

3.9 Identification of product - Equipment, assemblies, and parts shall be marked for identification in accordance with Standard MIL-STD-130.

3.10 Workmanship - The lampholder including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner, Particular attention shall be given to freedom from blemishes and defects, marking of parts and assemblies, thoroughness of soldering, cementing, welding, brazing, painting, wiring, and riveting, alignment of parts, and tightness of assemble screws and bolts.

3.11 Cleaning - The lampholder shall be thoroughly cleaned, and loose, spattered or excess solder, metal chips, and other foreign material removed during and after final assembly.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests - The inspection and testing of the lampholders shall be classified as follows:

(a) Preproduction tests (see 4.3)

(b) Acceptance tests (see 4.4)

4.3 Preproduction tests -

4.3.1 Preproduction test sample - The preproduction test sample shall consist of one lampholder, one lamp, (see 3.4.1) and one filter representative of the production equipment. It shall be tested in a place and manner designated in the contract, purchase order, or invitation for bids, to determine conformance with the specification. The approval of the preproduction sample by the activity concerned shall not relieve the contractor of his obligation to supply equipment conforming to this specification. Any changes or deviations from the preproduction sample shall be approved by the contracting officer. (see 6.2).

4.3.2 Tests - Preproduction tests shall consist of all tests specified under 4.5

4.4 Acceptance tests - Acceptance tests shall consist of the individual tests and sampling tests.

4.4.1 Individual test - Each lampholder shall be subjected to examination of product as specified in 4.5.1.

4.4.2 Sampling tests -

4.4.2.1 Sampling - One lampholder shall be selected at random from each lot of 100 or fraction thereof, and subjected to the assembly test and operation test specified in 4.5.2 and 4.5.3, respectively.

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4.4.2.1.1 Lot - A lot shall consist of lampholders manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

4.4.3 Rejection and retest - When one or more items from a lot fail to conform to the specification, acceptance of all items will be withheld until the extent and cause of failure are determined, and appropriately corrected.

4.4.3.1 Individual tests may continue - For production reasons, individual tests may be continued pending the investigation of a sampling test failure. However, final acceptance of the entire lot shall not be made until it is determined that the lot conforms to this specification.

4.4.3.2 Defects in items already accepted - The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.5 Tests -

4.5.1 Examination of product - Each lampholder shall be examined to determine compliance with the requirements specified herein with respect to materials, workmanship and marking.

4.5.2 Assembly - The lampholder shall be tested with the specified lamps and filter to determine that the design and construction is satisfactory. When assembled, there shall be approximately a 1-inch space between the lamp and the filter.

4.5.3 Operation - The lampholder shall be operated for 24 hours at room temperature (approximately 25° C.) with the specified lamps (see 3.4.1) and filter (see 3.5.8) and at rated amperes. There shall be no evidence of smoking, charring, softening, or other damage to any component of the light.

4.5.4 Color and transmission - The filter shall be tested to determine if the color and transmission are in accordance with Specification MIL-C-25050 for type + aviation colors. This inspection shall be made after the filter has been installed on the lampholder and the lampholder has been operated with the specified lamp for 1 hour at room temperature (approximately 25° C) at the rated amperes.

4.5.5 Lamp position - The lampholder complete with lamp shall be tested to ascertain that it is correctly positioned and aimed. The lampholder shall be checked to determine that aiming of the lampholder corresponds to the protractor scale. (See sheet 6 of MS90303).

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4.5.6 Dielectric test - The lampholder shall be given an applied 60-cycle ac voltage test of 1,000 volts for 1 minute between each terminal of the terminal block and the lampholder shell with the lamp socket and flexible leads connected in place. The terminal opposite to the one being tested shall be grounded to the shell. The terminal block shall be subjected to 5,000 volts dc to ground. After resting on a wet surface for 6 hours, insulation resistance of less than 2,000 megohms shall be cause for rejection.

4.5.7 Fungus resistance - Test in accordance with 4.8.1 of MIL-E-5272.

4.5.8 Salt spray - To insure compliance with applicable portions of 3.3.2 and 3.3.3, tests shall be conducted to conform to 4.6 of MIL-E-5272.

4.5.9. Temperature - The light shall be tested at 65°C in accordance with paragraph 4.1.2 and -54°C in accordance with paragraph 4.2.1 of MIL-E-5272.

4.6 Packaging, packing, and marking - Preparation for delivery shall be examined for conformance with section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging -

5.1.1 Level A - The lampholders shall be preserved in accordance with method IA of Specification MIL-P-116, without preservation, and packed in containers conforming to Specification PPP-B-566, PPP-B-665, PPP-B-636, or PPP-B-676. For domestic shipment, containers conforming to Specification PPP-B-636 may be used. Unit quantities shall be as specified by the procuring activity.

5.1.2 Level C - The lampholders shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing -

5.2.1 Level A - Lampholders preserved and packaged in accordance with 5.1.1 shall be packed in exterior-type shipping containers conforming to Specifications PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-636, MIL-B-138, or MIL-B-10377. As far as practicable, exterior shipping containers shall be of uniform shape and size, of minimum cube and tare consistent with the protection required, and contain identical quantities. The gross weight of each pack shall be limited to approximately 200 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Containers shall be provided with a case liner conforming to Specification MIL-L-10547 and shall be sealed in accordance with the appendix thereto. The case liner will not be required when the unit, intermediate, or exterior container conforms to Specification PPP-B-636 and is sealed at all joints and seams, including manufacturer's joint, with tape conforming to Specification PPP-T-60.

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The inspections set forth in this specification where such inspections are deemed

5.2.2 Level B - Lampholders preserved and packaged in accordance with 5.1.1 shall be packed in domestic-type exterior shipping containers conforming to Specification PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-636, or MIL-B-10377. Exterior shipping containers shall be of minimum cube and tare consistent with the protection required. As far as practicable, exterior shipping containers shall be of uniform shape and size and contain identical quantities. The gross weight of each pack shall be limited to approximately 500 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. When fiberboard containers are used, the fiberboard shall conform to table titled "Requirements for types I and II, class 1, domestic service fiberboard," of Specification PPP-B-636.

5.2.3 Level C - Packages which require overpacking for acceptance by the carrier shall be packed in exterior-type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall conform to Uniform Freight Classification Rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 Physical protection - Cushioning, blocking, bracking, and bolting, as required, shall be in accordance with Specification JAN-P-100, except that for domestic shipments, waterproofing requirements for cushioning materials and containers shall be waived. The drop tests of Specification JAN-P-100 shall be waived when preservation, packaging, and packing of the item are for immediate use or when drop tests of Specification MIL-P-116 are applicable.

5.4 Marking - In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use - The lampholders covered by this specification are intended for applicable use with a wheels-up warning system for aircraft over airports, and other Navy applications.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) The quantity, MS part number and type number of the lampholder desired (see 1.1).

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(c) Whether filters are to be furnished; also specify color (see 3.5.8)

(d) Whether lamps are to be furnished with the lampholders (see 3.4.1)

(e) Size of cord (see 3.5.6)

(f) Applicable level of packaging and packing (see 5.1 and 5.2)

(g) Where the preproduction test sample should be sent (see 4.3.1)

6.3 Provisions for preproduction testing - Manufacture of any additional lampholders on a particular contract should be suspended until the samples submitted for the preproduction tests are pronounced satisfactory by the procuring activity. When a contractor is in continuous production of these lampholders from contract to contract, submission of further preproduction samples on the subsequent contracts may be waived at the discretion of the procuring activity. Approval of preproduction samples or the waiving of preproduction tests does not preclude the requirements of acceptance testing.

6.3.1 It is understood that the lampholder supplied under contract or order will be identical with the corresponding preproduction sample in design, construction, quality, material, workmanship and method of manufacture. Evidence of unauthorized change may constitute cause for rejection.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-L-81015(Weps) LAMPHOLDER, PAR 46/64, AVIATION LIGHTING

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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Bureau of Naval Weapons (RREN-5)
Department of the Navy
Washington, D. C. 20360

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